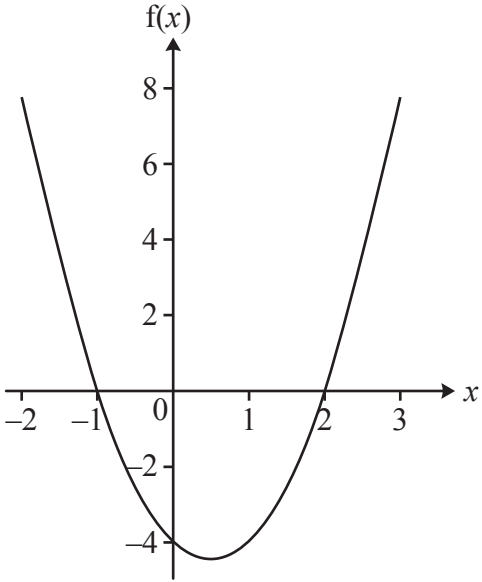
12 The diagram shows the graph of $f(x) = k(x-p)(x-q)$ where k, p and q are constants. The graph passes through the points (–1, 0), (0, –4) and (2, 0).



(a) Find $f(x)$ in the form $ax^2 + bx + c$. **[3]**

A cubic curve has gradient function $f(x)$. This cubic curve passes through the point (0, 8).

(b) Find the equation of the cubic curve. **[4]**

(c) Determine the coordinates of the stationary points of the cubic curve. **[3]**